DL Day 4

30 April 2026 05:58 PM

CNN

→ Convolutional Neural Network

↳ Image data → millions of pixels

→ Visual patterns → Edges, Shapes, Colors, Objects, faces etc

ANN → Structured data → Table

100×100 = 10,000 pixels → Imp weights training

Image → ANN → No

pixels → ▦

CNN → detect image patterns

→ learn step by step

↳ Face unlock

Self driving Cars

CLTV

Too many Cats → CNN → This is Cat

Cat

Flow

Input image → Layer Convolution → Layer Pooling → Feature Extraction → O/P

ACF

N

1) Input layer

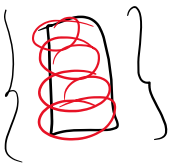
→ Image → CNN

2) Convolution layer

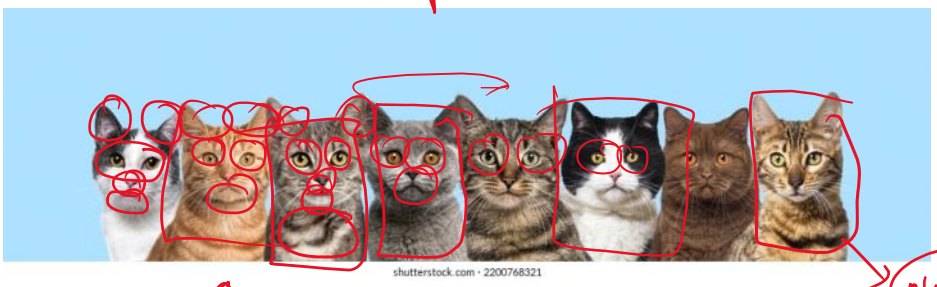
... images and detect useful patterns

a) Convolution layer

Scan the input image and detect useful patterns



Input



New data

Convolution → produce output values

o/p

Cat

Activation Function (RELU)

↳ Rectified linear unit

Imp → Pooling

Keep most Imp info

Human → Hawa Car (1) (1)

Imp

Car

Conv

Pool → Car ✓

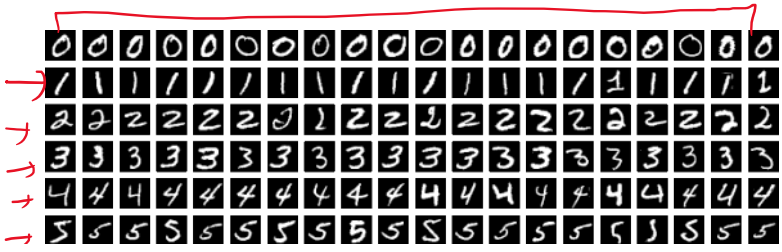
CL → RELU → PL

max Pooling → 5

Avg pooling → $\frac{1+5+3+2}{4} = \frac{11}{4}$ (2)

[1, 5, 3, 2] ✗

Image → CL → RELU → PL → CL → - - - - - - → o/p

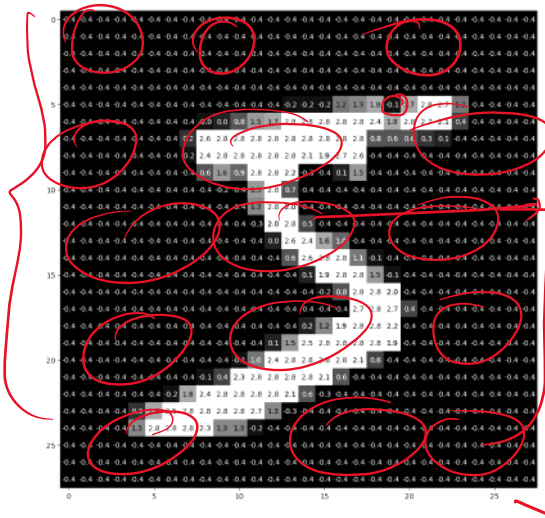


(0)

→ Actual data → 6

→ predict the digit

→ 6



→ label → 5

→ Pooling → +ve

5

→ actual data
it is 5

What is th

→ input → Conv → Pooling →
Predict → o/p → 5